

hdh | an **agent-first EHR**
notes in · orders out · nothing guessed

```
$ hdh comprehend --file note.txt --apply  
confirmed hypertension already on the problem list  
updated controlled well treated → flag set  
new HbA1c lab order, due in 90 days  
review unmapped term NOT charted – sent to a human
```

SNOMED CT · ICD-10-CM · LOINC · RxNorm

10,000 synthetic patients · zero PHI · MIT

Domain performance is an orchestration problem now, not a training one.

The assumption was that clinical AI meant fine-tuning a small model on clinical text.

Frontier reasoning models already carry the domain knowledge. What they lack is **this patient's chart, your terminologies, and somewhere legitimate to put the answer.**

Supply those three through tools and orchestration and you get domain-grade behaviour without training anything — and the behaviour holds when you swap the

model, because the parts that must not be wrong were never in the weights.

THE SHIFT

The hard part moved out of the weights

hdh is model-agnostic on purpose. The same orchestration runs on a frontier reasoning model or a considerably smaller one, because grounding, coding, reconciliation and refusal live in **deterministic code** rather than in a checkpoint. The model's job is narrow and it never widens: *this span is a problem*, *this span is its dose*. It never picks a code.

That split is what makes the claim testable rather than rhetorical. Swapping the model changes how well spans are *found*; it cannot change what gets *charted*. A weaker model finds fewer mentions and sends more of them to a human — it does not quietly invent a diagnosis, because nothing in the path lets it.

FREE TEXT IN

A clinician talks. The chart updates.

No forms, no template, no structured entry. A dictated sentence — or, once speech capture lands, the same sentence spoken at the bedside — becomes reconciled, coded chart entries. Seven passes run between the text and the verdicts below.

"Patient with h/o type 2 diabetes came in with an HbA1c over 7. Continued metformin, added Januvia, repeat HbA1c in 90 days."

confirmed	type 2 diabetes	already on the problem list
updated	controlled	HbA1c > 7 → flag cleared
new	Januvia 25 mg	prescription, coded to RxNorm
new	HbA1c	lab order, due in 90 days
review	unmapped term	NOT charted – sent to a human

The problem list is reconciled rather than duplicated. Orders and prescriptions are coded at the moment of writing, not by a coder three days later.

GROUNDING

The interesting part is where it refuses

Hallucination in a chart is not a bad sentence — it is a row that looks exactly like a real one. Orchestration is what makes refusal possible at all, because a deterministic layer can decline where a model will oblige. These three took longer to build than everything around them.

It never invents a lab result

"HbA1c over 7" does not become a lab result. No specimen, no method, no reference range, no performing lab — the value is being *referred to*, not reported. File it and the chart holds two kinds of row that look identical: one from an instrument, one from a recollection. Results arrive from a lab, matched to the order that asked for them.

It never guesses a drug strength

"Start metformin" codes to the ingredient, not to a 500 mg tablet. The note said what it said. A drug is where a confident guess does the most harm, so the coding stops at the deepest level the evidence actually supports.

It never charts what it cannot code

Unresolvable mentions go to a review queue for a human, with the sentence they came from. The alternative is a chart full of confident approximations that nobody notices — and unlike a wrong sentence, a wrong coded row is acted upon.

6 → 1

Confident *wrong* answers on a stress corpus of real clinical phrasings, and the fixes came from measurement rather than intuition. "Type 2 diabetes" was returning "Cataract due to diabetes mellitus type 2" at 100% confidence, because nothing in the ranking penalised a term for saying **more** than the note did.

Most clinical AI writes *into* the chart. This works *from* it.

Ambient scribes and assistant side-panels are genuinely useful, and they stop in the same place: they produce prose for a human to file, or they answer questions about the schedule. They hold a transcript rather than a chart, so they cannot reconcile against a problem list they do not have — and what they emit is text, so nothing downstream can act on it.

	TYPICAL EHR AI ASSISTANT	AGENT-FIRST EHR
Sees	This encounter, or a transcript of it	The whole chart, its history, and the panel around it
Produces	Prose a human then files	Reconciled chart rows, coded orders, an audit entry
Coding	A coder, afterwards	SNOMED CT, ICD-10-CM, RxNorm, LOINC at write time
When unsure	Produces plausible text anyway	Refuses, and queues it for a human with its source sentence
Reach	Documentation, or scheduling	Documentation → orders → results → care gaps → care plan
Corrections	Edit the text	Amend or void, reason required, never deleted

One agent, the whole clinical day

Because the agent holds the chart rather than a transcript, documentation stops being the end of the interaction and becomes the start of it. Everything below reads the same chart the note just changed.

- **Orders, and the results that answer them**

A note proposes drafts; releasing is a human act. Results return from a lab and are matched to the order that asked — a result nobody ordered is never filed.

- **Care gaps that move when the note does**

"*Well treated hypertension*" clears a flag; "*h/o type 2 diabetes*" puts a problem on the chronic list. The morning's dictation changes the afternoon's outreach list, because they are one system.

- **Predictive risk across the panel**

Probability of an urgent visit or a critical result within 180 days, learned from chart patterns and surfaced to the agent as one more tool it can cite.

- ⦿ **Care plan generation** NEXT

Goals, interventions and review dates grounded in the problem list and the guidelines behind it — a proposal a clinician approves, refusing on the same terms as everything else.

- ⦿ **Speech capture** NEXT

The pipeline already takes free text, so audio is a front door rather than a rebuild: the same sentence spoken at the bedside reaches the same seven passes.

- ⦿ **Type, talk, or upload** NEXT

A React / Next.js front end that takes whatever the clinician has: typed text, dictation, an outside note as a PDF, a lab report, an image carrying a radiologist's findings. Each new input is a new *front door* — never a new set of rules. Everything meets the same gate, is coded where the evidence supports it, and goes to a human where it does not.

What is deliberately missing

A real EHR is mostly the things below, and hdh has almost none of them. That is a choice, not a gap in the roadmap — none of it is difficult in the interesting sense, and all of it is **compliance-shaped** rather than research-shaped.

🕒 Scheduling

Appointments, slots, templates, waitlists, recalls. Thoroughly solved elsewhere, and it teaches nothing about agentic AI.

👤 Clinician identity

Profiles, roles, credentials, licensure, supervising relationships. hdh carries a thin provider record and stops there.

🔒 Access control

Authentication, RBAC, break-glass, consent, tenancy, encryption at rest and in transit, retention.

These belong to a separate workstream — one that goes through HITRUST and HIPAA review, at that discipline's pace and under that scrutiny. Bolting a half-version onto a learning repository would teach exactly the wrong lesson: that a control environment is a feature list. It is not, and pretending otherwise is how systems that handle real patient data go wrong.

An MIT-licensed way to learn agentic AI in healthcare

Not a product, not a validated clinical tool, and not seeking to be either. It is a complete, readable worked example of applying agentic AI to clinical work — with the design arguments written down beside the code, including the ones that turned out wrong and what the measurement said instead.

Every patient in it is synthetic, which is the point: you can run it, break it, chart nonsense into it and read exactly why it refused, without anyone's data being anywhere near it. Clone it, point it at PostgreSQL, and talk to it.

github.com/arsalanam/hdh

MIT licensed · open source · Python

10,000 synthetic patients · 165,000+ visits · 777,000+ lab results · zero PHI by construction. A proof of concept, not a product, and not a validated clinical tool.

[SNOMED CT](#) · [ICD-10-CM](#) · [LOINC](#) · [RxNorm](#)